A shuttle, rethought.

KESTREL

Compact for flight. Expansive in orbit.



Illustrative concept rendering. Geometry, deployed radiator area and mechanisms are not engineering drawings.

CREW	USEFUL CARGO UP	ORBITAL MASS
6 people	3 tonnes	25 tonnes

A reusable lifting body rides above a separate launch vehicle, unfolds solar and cooling surfaces in orbit, then returns people and valuable cargo to a runway. The design concentrates on routine low Earth orbit transport and return logistics.

Study status: pre-Phase A concept and screening calculations. This brief selects an architecture, allocates mass and energy, and identifies the experiments needed to make the concept real. It does not establish flight safety, certified performance, cost, or a construction-ready design.

Design around a real job.

The mission defines the vehicle. Every number below is a proposed requirement or allocation, unless explicitly identified as heritage.

Baseline target	Meaning / boundary
Destination	400 km circular low Earth orbit; launch directly into the destination orbital plane.
Transport	Six crew plus 3 t useful up-cargo; up to 2 t useful return cargo, within a separately qualified landing mass and center-of-gravity envelope.
Duration	Seven days independent operation. Longer docked stays require qualified station utilities, seals, propellant storage and periodic health checks.
Envelope	Approximately 15 m length and 8 m aerodynamic span. Solar deployment grows the span beyond 20 m.
Recovery	Autonomous unpowered runway landing; crew can command aborts and supervise. Runway length, speed, crosswind and cross-range remain unclosed.
Payload interface	One central cassette; useful payload excludes the cassette carrier already allocated in dry mass. Volume, loads and access remain interface trades.
Initial launch system	Recovered booster, expendable upper stage with controlled disposal, reusable orbiter. Full-system reuse is a later trade, not the baseline.

What earns the wings

Runway return is valuable when sensitive instruments, biological samples, reusable equipment and people need controlled handling near a prepared facility. If the job is only inexpensive up-cargo with little return value, a capsule or cargo-only launcher may win. That comparison remains open.

A deliberate reduction in scope

The original Shuttle carried a much larger payload bay and supported missions this vehicle cannot replace. NASA lists its orbiter at roughly 76-79 t empty and 37 m long. KESTREL targets 18 t dry and a routine 3 t cargo task; smaller overhead is partly purchased by giving up large-satellite transport. [1]

Heritage: NASA Shuttle facts [1]; NASA HL-20 architecture research [2]. KESTREL is an original synthesis of established concepts; no patent novelty is claimed.

Keep ascent hardware off the shuttle.

Use a staged rocket to provide almost all ascent velocity. The orbiter carries only the propulsion needed for rendezvous, separation, entry targeting and attitude control. A top-mounted configuration is the starting point, subject to coupled aeroelastic, launch-load and abort analysis.

Element	Baseline choice	Why / tradeoff
Booster	Reusable liquid-propellant stage; methane/oxygen is a candidate	Aim for recoverable hardware and maintainable engines. Methane is not automatically more energy-efficient than hydrogen.
Upper stage	Separate expendable stage initially; propellant and launcher selected by integrated trade	Avoid adding orbital recovery mass before the transport mission closes. Reserve the stage's own disposal propellant.
Orbiter	Reusable lifting body, electric control actuation where justified	Retain the return cabin, thermal protection and landing system; omit the large ascent engines.
Launch escape	Dedicated whole-vehicle separation/escape architecture, not yet sized	Must demonstrate clearance, loads, attitude control and recovery over the required ascent envelope before crew flights.

Architecture alternatives

Option	Decision at this stage
Capsule on reusable booster	Keep as the economic and safety comparator. Likely simpler if runway down-cargo is not valuable.
Winged orbital second stage	Retain as a future option; returning ascent tanks and engines may erase the smaller-shuttle benefit.
Airbreathing two-stage system	Research option. Precoolers and engine-mode transitions add integration and thermal demands.
Single stage to orbit	Exclude from the baseline. At an illustrative 9.3 km/s and 380 s specific impulse, only 8.2% of initial mass remains for all hardware, reserve and payload. [3]

ESA's SABRE precooler ground test is component evidence; its later system study selected a two-stage architecture. Neither establishes a flying orbital spaceplane. [4,5]

Twenty-five tonnes is a budget.

Bottom-up engineering has not yet validated these subsystem allocations. The 20% dry-mass allowance is a concept reserve, not evidence of maturity.

Allocation	Mass
Structure, pressure shell and aero surfaces	5.2 t
Thermal protection	2.2 t
Landing gear and flight controls	1.1 t
Orbital propulsion hardware and tanks	1.2 t
Life support hardware	1.1 t
Solar, batteries and electrical distribution	1.5 t
Thermal control hardware and working fluids	0.7 t
Avionics and communications	0.5 t
Docking, cassette interface and inspection arm	1.0 t
Cabin interior and seats	0.5 t
Basic dry estimate	15.0 t
Growth allowance: 20% of basic dry estimate	3.0 t
Total dry	**18.0 t**
Crew, suits/personal gear and mission consumables	1.0 t
Useful cargo at insertion	3.0 t
Orbital propellant, including 0.4 t held allocation	3.0 t
Orbital insertion total	**25.0 t**

Additional at liftoff: launch adapter, escape motors and any shroud or protective structures. Their masses are unresolved and are not hidden in the 25 t orbital figure. An unflown NASA HL-42 concept allocated about 7.6 t to abort motors and adapters on top of its 21.1 t on-orbit mass. That is a warning about scale, not a KESTREL estimate. [6]

Mass-growth rule: with the same 25 t insertion interface, each extra tonne beyond the budgeted 18 t dry mass removes one tonne of cargo. An additional 3 t consumes the entire cargo target. Above that, resize the launcher/vehicle or change the mission.

Efficiency starts with less mass.

CALCULATED MECHANICAL-ENERGY INCREMENT

9.20 MWh per added tonne

Surface at rest to 400 km circular orbit; spherical, nonrotating Earth. Propulsion and trajectory losses are excluded.

Using Earth's gravitational parameter and a 6,371 km mean radius, the state-energy change is **mu/R - mu/[2(R+h)] = 33.13 MJ/kg**. For the 25 t orbital vehicle, that is about **230 MWh of mechanical energy**. It is not a launch-fuel estimate and cannot be divided by solar-panel output to describe an electric launch.

Four separate efficiency claims

Measure	What KESTREL changes	What must still be measured
Transport overhead	A compact orbiter removes mass associated with the original large cargo mission and ascent engines.	Same crew, useful cargo, return manifest, orbit and service duration.
Orbital energy	Solar/battery power replaces routine fuel-cell reactant consumption.	Array, battery, cooling and maintenance mass over actual use.
Hardware reuse	Booster and orbiter return for inspection and reuse.	Recovery propellant, discarded upper stage, refurbishment and achieved lifetime.
Lifecycle impact	Design for accessible modules and fewer replacements.	Propellant production, launch emissions, electricity, manufacturing and ground operations.

What I would measure before claiming a win

Report primary energy per completed six-person/3 t delivery and 2 t return mission; useful up- and down-cargo per flight; occupied-seat utilization; labor hours per turnaround; replaced hardware mass; and completion reliability. Keep cost and energy as separate results. The study does **not** establish a percentage improvement over Shuttle or a modern capsule.

A future orbital workshop can reduce repeated transport: leaving a tool set of mass m in orbit avoids roughly (N - 1)m of repeated up-mass over N visits, before counting its support module, servicing, storage and disposal. This benefit requires enough visits to repay that overhead.

Enough maneuvering for the mission.

The reference propulsion case uses an assumed **315 s effective specific impulse**, representative of conventional storable bipropellant performance. This is a modeling assumption, not a selected engine. Long-duration storage is attractive; toxicity and ground handling are real costs. A lower-toxicity substitute requires its own mass and performance trade.

ROCKET-EQUATION SCREENING RESULT

339 m/s available

25.0 t initial / 22.4 t final; 2.6 t burned, 0.4 t held for attitude use, residuals and unusable propellant. [3]

Main-maneuver allocation	Delta-v
Rendezvous, phasing and docking approach	120 m/s
Deorbit targeting	110 m/s
Trajectory dispersions and operational allowance	50 m/s
Contingency allowance	50 m/s
Allocated total	**330 m/s**
Unallocated arithmetic remainder	9 m/s

This is an allocation to be tested against a trajectory. The launch stage must deliver the proper orbital plane and insertion state. Rendezvous timing can consume more than the allowed 120 m/s; if so, change the launch window, propulsion mass or mission.

The boundaries are sharp

An ideal retrograde burn from circular 400 km to a 50 km perigee is about **102 m/s**. Just a **1 degree plane change costs 134 m/s** at this altitude. KESTREL is a same-plane transport, not an orbital touring vehicle. Entry targeting also has heating, range and landing-site constraints that this two-body calculation omits.

The upper stage uses its own budget for controlled disposal. Orbiter reserves are not silently spent on it. The 0.4 t held propellant is separate from the 50 m/s main-engine contingency allowance.

Unfold a working power station.

25 kW means reference sunlit output delivered to the bus. It does not mean 25 kW continuous power through eclipse.

Design input / result	Screening value
Nominal spacecraft electrical load	8 kW continuous; 15 kW short-duration total peaks
Reference solar output	25 kW, under the assumed illumination and derating
Active photovoltaic area	81 m2; 1,361 W/m2 sunlight x 29% cells x 0.78 derating
400 km orbit / maximum eclipse	92.4 min period / 36.0 min eclipse
Eclipse energy at 8 kW	4.81 kWh delivered
Minimum sunlit generation for energy balance	13.7 kW, assuming 90% storage round-trip efficiency
Battery allocation	40 kWh nameplate, split into two isolated 20 kWh packs
End-of-life delivered battery energy	23.6 kWh: 80% capacity x 80% depth of discharge x 92% discharge efficiency

The margin has a job

The gap between 13.7 and 25 kW supports imperfect pointing, charging, degradation uncertainty and scheduled payload peaks. Actual energy margin depends on orbit, orientation, shadowing and array condition. Peak-power duration and battery discharge rate still need qualification.

Make the deployed shape honest

About 40 m2 of active cells per side creates a much larger orbital silhouette than the 8 m stowed aerodynamic span. Choose articulated rigid panels with qualified hinges, latches, cable flexures and stow sensors. A deployed solar technology is not automatically a repeatably retractable one: NASA's 2017 ROSA experiment could not retract and was jettisoned. [7]

One surviving battery pack gives about 11.8 kWh after the stated derating, or 2.9 hours at an assumed 4 kW emergency electrical load. This is not a demonstrated safe-return duration. Entry needs a verified essential-load list and a bounded stow-to-ground-power interval.

Solar irradiance reference [8]. Cell efficiency, derating, power loads, battery capacities and margins are concept assumptions.

Cooling is the hidden size driver.

Solar panels face the Sun; radiators need favorable views of cold space. Use **separate deployed surfaces**, with shielding and attitude rules that respect the docking corridor. A large supply of electricity is useful only when the spacecraft can reject the resulting waste heat. [9]

Screening circuit	Assumptions	Effective area
Warm electronics / payload	8 kW; 310 K; emissivity 0.85; 150 W/m2 environmental penalty	27 m2
Cool cabin / dehumidification	4 kW; 285 K; same simplified penalty	24 m2
Total / target allocation	Idealized net radiative balance; full orbital thermal model still required	**51 m2 / 55-65 m2**

The 12 kW rejection case includes electrical conversion, crew and environmental allowances; it is not the 25 kW solar rating converted directly into cabin heat. Actual cabin cold surfaces, heat exchangers, pumps and radiator temperatures require a coupled design. Both sides of a panel count only if both have useful radiative views.

Two distinct thermal problems

In orbit: fold-out, multileaf radiator panels with isolated warm/cool circuits, redundant cooling paths and leak detection. Packaging 55-65 m2 of effective radiator area is a major unresolved issue; the concept art does not resolve it.

During entry: stow every exposed appendage, then use thermal storage and/or expendable cooling until ground services connect. An 8 kW heat load for 90 minutes requires **43.2 MJ** of heat handling. This is an energy allocation, not a sized cooler. Shuttle radiator operations provide heritage for changing cooling modes before landing. [10]

A heat shield designed to be inspected

Use toughened ceramic acreage tiles, TUFROC-derived high-temperature elements at suitable nose/leading-edge locations, and insulating blankets in cooler zones. NASA reports flight-tested reusable TUFROC material to roughly 1,590 C; that temperature is not a qualified limit for KESTREL's shape, joints or attachments. [11]

Aerodynamic heating, gaps, impact damage, repeated-cycle durability, water absorption, tile bonding and panel replacement tolerances must be verified. This study claims neither self-healing protection nor a solved reentry trajectory.

Three features worth building.

01 The orbital transformation

The compact shuttle unfolds into a solar-powered workspace, with its mission cassette between separate power and cooling structures. This creates a striking shape for a physical reason: sustained work in space needs exposed collection and rejection area. The work bay supports robotic payload operations; an EVA airlock is not in the baseline.

Value: payload power and thermal services without a large consumable power supply. **Gate:** reliable deployment and stowage, docking clearance, contamination control and recovery from mechanism faults.

02 A recorded inspection before every return

A compact robotic arm scans the return surfaces with cameras and depth measurement, supported by impact and temperature sensors. The system compares the current surface with its recorded baseline and highlights damage for qualified review. Shuttle's inspection boom provides operational precedent. [12]

Value: earlier damage awareness and traceable maintenance. **Gate:** demonstrate coverage, detection thresholds and false-negative rates, including the belly, hinges and hidden joints. Inspection discovers a problem; it does not repair it or guarantee survivability.

03 The mission arrives as a cassette

A removable carrier holds cargo restraints or a powered experiment package, with controlled mechanical, electrical, data and cooling interfaces. Prepare the next cassette while the shuttle is being inspected. Return-sensitive equipment can remain secured and powered through landing and ground hookup.

Value: reusable fixtures and parallel preparation. **Gate:** each load must pass mass, center-of-gravity, venting, fire, thermal and structural checks. A fast swap is a turnaround objective, not an established flight cadence.

The later capability: leave the workshop in orbit

At sufficient flight rate, move heavy tools and long-lived equipment to a compatible orbital work module. The shuttle carries people, feedstock and finished products, then returns only the manifest that benefits from landing. An additional module, robotic transfer and sustained operations all carry their own mass and cost.

These capabilities combine existing ideas in a proposed vehicle. They are distinctive design choices, not claims of unprecedented invention.

Design the return before the launch.

Mission step	Required behavior
Prepare	Inspect the vehicle; install a verified cassette; upload its mass/CG record; verify reserves, destination and landing alternatives.
Launch / insert	Use a compatible launcher and demonstrated escape envelope. Separate cleanly; reserve upper-stage disposal resources independently.
Deploy / rendezvous	Verify cooling and solar deployment before high-load operation. Approach within proven plume, collision and docking-clearance limits.
Dock / work	Use destination-qualified interfaces; track consumables and return reserves continuously. Restrict large power loads when thermal capacity is limited.
Inspect / commit	Complete TPS coverage, assess damage against qualified limits, confirm weather and landing opportunities, and prove appendage stow.
Enter / land	Use essential power and alternate cooling. Fly a qualified entry corridor and autonomous approach; connect ground services after landing.
Recover / turn around	Safe hazardous propulsion and batteries; verify cabin conditions before cargo access. Replace failed modules; retain life-cycle records.

Crew support that matches seven days

Use vacuum-regenerated carbon-dioxide removal, stored water and breathing gases, emergency suit support and replaceable filters. Orion provides heritage for regenerative scrubbing with stored consumables. If captured humidity is vented, do not also count it as recovered water. [13]

The 1.0 t mission allocation provisionally reserves about 0.66 t for six occupants with suits/personal gear and 0.34 t for consumables and reserves. It requires a detailed manifest and loss model; seven-day endurance is not yet closed. Full urine/water recycling is deferred until mission duration justifies its hardware and power.

Rescue-compatible, not rescue on demand: an IDSS-compatible docking interface must also pass destination-specific pressure, electrical, data and operations checks. Rescue needs a ready vehicle, favorable phasing, life-support time and empty return seats. A full six-seat cabin has no spare rescue seats. [14]

Prove the difficult parts first.

Gate	Evidence required before proceeding
Mass and packaging	Integrated CAD, equipment layouts, wiring/fluid runs and subsystem mass estimates. Close radiator/solar stowage, usable cabin and cargo volume; retain growth margin.
Launch and escape	Coupled loads and aeroelastic analysis; pad, high-load ascent and high-altitude escape tests. Demonstrate separation, stable recovery and survivable crew loads.
Aerodynamics and entry	Wind-tunnel/CFD correlation, aeroheating, trim and cargo-CG envelopes; arc-jet TPS tests; landing and crosswind evaluation.
Power and thermal	Thermal-vacuum orbital cycles with real loads; worst-case eclipse; battery fault containment; failed-loop and stowed-descent cooling cases.
Moving structures	Life-cycle deployment/stow tests in representative thermal/vacuum conditions; jam detection, alternate actuation and qualified fail-safe outcomes.
Flight software	Independent safety assessment; hardware-in-the-loop guidance and fault injection; verified docking inhibits and degraded landing modes.
Reuse and economics	Uncrewed orbital return, instrumented inspections and repeated flights; record labor, replaced parts, actual energy and achievable dispatch reliability.

A development sequence with exit criteria

First close the coupled sizing model and retire the highest-risk mechanisms on the ground. Then fly atmospheric approach/landing articles and TPS/entry demonstrators. Conduct uncrewed orbital cargo missions and repeated reuse demonstrations. Admit crew only after the launcher, escape envelope, return systems and operating procedures satisfy human-rating requirements.

No calendar, development price, per-seat cost or numeric reliability is promised here. Those require a funded program definition, supplier estimates, a flight-rate model and probabilistic safety analysis.

DESIGN-REVIEW VERDICT

Promising, with explicit gates.

Proceed with integrated sizing and abort/thermal trades. Do not freeze the external shape or sell the target specifications as established capability.

What this study establishes.

Established in this study	Still open
A coherent mission and architecture: compact crew/cargo orbiter on a separate staged launcher.	Launcher choice, launch adapter/escape mass, aerodynamic compatibility and insertion performance.
A transparent 25 t orbital mass allocation with 3 t dry-mass growth allowance.	Subsystem estimates, packaging, structural margins and whether 6 crew + 3 t cargo truly fit.
Consistent first-order delta-v, orbital energy, solar, battery and radiator calculations.	Coupled trajectories, detailed thermal balances, plume environments, fault cases and operational reserves.
Three concrete differentiators with testable utility.	Demonstrated mechanism life, inspection reliability and cassette turnaround time.
A route to test energy and operating-effort improvements.	A measured advantage against Shuttle, modern capsules or other reusable launch systems.

Engineering choices I would protect

Keep the main ascent engines on the launcher. Keep routine mission size modest. Reserve volume and mass for real radiators. Treat escape as a first-order architecture requirement. Make damage inspection and maintainability design inputs. Build a useful cargo return system before adding more destinations or exotic propulsion.

Choices I would allow the evidence to change

The exact shape, span, launcher fuel, solar power level, radiator temperature, propellant chemistry and crew/cargo combination are all negotiable. If radiator packaging or escape mass breaks the 25 t interface, reduce payload power or cargo, enlarge the vehicle/launcher, or compare again with a capsule. The concept should survive through better choices, not through optimistic bookkeeping.

Reproduce the arithmetic

The accompanying **kestrel-sizing.py** uses only Python's standard library. Run it to regenerate the numerical results. The accompanying JSON records assumptions and calculated outputs. It is a screening model, not a flight dynamics, aerothermal, life-support or certification tool.

Artwork was generated with the built-in image generation tool from a prompt included with the deliverables. It communicates form and deployment intent; it does not validate dimensions, panel area, structural load paths or component clearances.

Sources and assumptions.

Primary aerospace references accessed 8 October 2026. The links support heritage and physics; the proposed vehicle targets are this study's assumptions.

[1] NASA - A Versatile Vehicle / Shuttle facts

Original orbiter dimensions, empty-mass range and mission scope.

[2] NASA - HL-20

Compact lifting-body crew-return architectural heritage.

[3] NASA Glenn - Ideal Rocket Equation

Mass-ratio and specific-impulse relationship. KESTREL inputs and calculations are this study's own.

[4] ESA - Air-breathing engine precooler achieves Mach 5 performance

Component ground-test scope; not a complete orbital propulsion demonstration.

[5] ESA - System study results for SABRE-powered reusable launcher

Two-stage architecture selection in the referenced study.

[6] NASA - Access to Space Study, January 1994

Figure 21, printed page 31 / PDF page 41: historical, unflown HL-42 mass allocation. Not a validation of KESTREL.

[7] NASA - Crew preps for solar array jettison and Dragon departure

2017 ROSA retraction outcome; motivation for a separately tested stowable system.

Sources, continued.

Primary aerospace references accessed 8 October 2026. The links support heritage and physics; the proposed vehicle targets are this study's assumptions.

[8] NASA Goddard - Radiative transfer research

Reference solar irradiance near Earth used in array-area sizing.

[9] NASA - State of the Art: Thermal Control

Thermal balance, radiation and orbital environmental considerations; not KESTREL radiator sizing.

[10] NASA - STS-74 Space Shuttle Mission Report

Printed page 29 / PDF page 35: radiator cold soak through landing and transition to ammonia cooling before ground hookup.

[11] NASA - Thermal Protection Materials Branch: Reusable Materials

TUFROC material heritage, modularity and reported temperature capability.

[12] NASA - Behind the Scenes: Real-Time Shuttle Debris Analysis

Shuttle orbital TPS inspection precedent.

[13] NASA - Orion Crew Systems

Life-support and regenerative CO2/humidity-removal heritage.

[14] NASA - International Deep Space Standards

Interoperable interface standards, including docking. Destination-specific qualification is still required.

Common assumptions

Spherical nonrotating Earth: mean radius 6,371 km; gravitational parameter 3.986004418 x 10^14 m3/s2. Circular 400 km reference orbit. Worst geometric eclipse uses zero orbit-Sun beta angle and a point Sun. Standard gravity 9.80665 m/s2. Radiative screening uses the Stefan-Boltzmann relation; the assumed 150 W/m2 environmental penalty is not a solved view-factor model.

No direct quotation is used. Historical reports describe studies and hardware of their time; this brief does not adopt their past forecasts as current fact. Numerical model, independent architectural review and operations review were reconciled before preparation.